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About

Dr. Shichao Ma earned his PhD from the Li Ka Shing Faculty of Medicine at the University of Hong Kong, where he was recognised as an HKU Presidential PhD Scholar. He also holds an MSc in Computer Science from the University of Hong Kong and a BSc (First Class Honours) in Computer Science from King's College London. Currently, he serves as a Lecturer in the Department of Computer Science at Hong Kong Baptist University. His research interests include deep learning, medical AI, digital health, and related fields.

Research Interests

- Deep learning
- Digital Health
- Artificial Intelligence in Medicine
- Biomedical Data Science

Selected Publications

- **S. Ma,** Y.-H. Chan, E. C. H. Ngai, and J. W. K. Ho, Asynchronous and focal federated learning for skin lesion classification under local data scarcity and class imbalance, Computer Methods and Programs in Biomedicine. 272 (2025) 109073. https://doi.org/10.1016/j.cmpb.2025.109073.
- **S. Ma**, J. Chen, J.W.K. Ho, An edge-device-compatible algorithm for valvular heart diseases screening using phonocardiogram signals with a lightweight convolutional neural network and self-supervised learning, Computer Methods and Programs in Biomedicine. 243 (2024) 107906. https://doi.org/10.1016/j.cmpb.2023.107906.
- **S. Ma**, C.-Y. Yes, J. Xin, J.W.K. Ho, Artificial intelligence clinical applications of wearable technologies, in: V. Vardhanabhuti, K.-W. Kwok, J.Y.K. Chan, Q. Dou (Eds.), Machine Learning, Medical AI and Robotics, IOP Publishing, 2023: pp. 3-1-3–16. https://doi.org/10.1088/978-0-7503-4637-5ch3.
- **S. Ma**, X. Fang, Y. Yao, J. Li, D.C. Morgan, Y. Xia, C.S.M. Kwok, M.C.K. Lo, D.M.D. Siu, K.K. Tsia, A. Yang, J.W.K. Ho, StarmapVis: an interactive and narrative visualisation tool for single-cell and spatial data, Computational and Structural Biotechnology Journal. (2023) S2001037023000697. https://doi.org/10.1016/j.csbj.2023.02.023.
- G. Jia, H.-K. Lam, **S. Ma**, Z. Yang, Y. Xu, B. Xiao, Classification of Electromyographic Hand Gesture Signals Using Modified Fuzzy C-Means Clustering and Two-Step Machine Learning Approach, IEEE Trans. Neural Syst. Rehabil. Eng. 28 (2020) 1428–1435. https://doi.org/10.1109/TNSRE.2020.2986884.